Quantifying the Impact of Exchange Rate Volatility on Key Economic Sectors in Sri Lanka

for the Bachelor of Science Honours Degree in

Financial Mathematics and Industrial Statistics

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Abstract

The variations in exchange rates have a great impact on the economic performance of an emerging economy such as Sri Lanka. The past few years have experienced a lot of volatility in the Sri Lankan rupee that has impacted a wide range of industries such as tourism, exports, imports, and financial markets. This paper will examine how LKR/USD exchange rate fluctuations affect imports, exports and tourism earnings in 2015-2025. Monthly series were created based on weekly and annual data using secondary data provided by the Central Bank of Sri Lanka and the Department of Census and Statistics. The missing values were interpolated and the outliers were handled with logarithmic transformations.

The research design entails correlation, multiple regression, and time series prediction methodology to explore the relationship patterns between exchange rate, interest rates, inflation, GDP, and sectoral performance. Regression findings reveal that exchange rate and GDP have a significant effect on imports; exports are moderately sensitive to fluctuations in the currency and tourism earnings are slightly sensitive to macroeconomic stability as opposed to fluctuations in currency. Short-term sectoral predictions were done using time series trend analysis, decomposition, moving averages and smoothing by Winters. Based on projections, early 2025, the level of imports is expected to be USD 1,554 million, exports USD 1,097 million, tourism earnings USD 118-266 million.

These results place significance on the management of exchange rates as a way of maintaining stability on trade, and further on the macroeconomic policies to enable tourism resilience. By influencing the literature on exchange rate dynamics in emerging markets, the analysis can be useful in making practical implication on economic policy and investment decision-making in Sri Lanka..

Keywords: Exchange Rate, Imports, Exports, Tourism Earnings, Time Series Forecasting, Regression Analysis

1. Introduction and Research Background

1.1 Introduction

One of the major macroeconomic indicators is the exchange rate, which influences trade, investment, and the stability of the economy. Under an open economy, any fluctuation in the value of the local currency will have a direct impact on the domestic price of imported goods and/or exports, the attractiveness of foreign investment, as well as the purchasing power of consumers. In the case of developing nations such as Sri Lanka, where major aspects of the economy, such as tourism, exports, and imports, play a significant role in the economy, the volatility of exchange rates can have far-reaching consequences on businesses and policymakers.

Over the past years, Sri Lanka has been struggling with significant economic issues and faced such problems as the weak state of currency, the pressure of inflation, and the disturbances in international trade. These shocks have been exacerbated by externalities like changes in global commodity prices, geopolitical tension, and international financial crises. The volatility of the Sri Lankan rupee against the foreign major currencies, particularly the US dollar, has affected the competitiveness of local exports, the price of imports, and the balance of payments in general. The exchange rate movements have also influenced the tourism revenue, which is one of the main sources of foreign exchange in the country, since the prices of travel and accommodation change among foreigners who visit the country.

1.2 Importance of the problem

Exchange rate volatility has been an economic challenge of great concern in Sri Lanka, particularly since the recent crisis. The impact of fluctuation in the value of the Sri Lankan rupee on other economic sectors such as tourism, exports, imports, and the financial market can be very high. However, there is a lack of empirical evidence on the effects of these changes on particular economic performance. Policymakers and investors would not easily make informed decisions relating to the economic planning, trade policies, and investment without knowledge of these relations. This research aims to statistically examine the impacts of the change in the exchange rates on the selected sectors of the economy in Sri Lanka.

Although it is important, there are limited studies based on empirical research specifically examining the effects of exchange rate fluctuations on the performance of key economic sectors in Sri Lanka. Literature is usually biased towards more macroeconomic indicators or analyzes trade and inflation separately, rather than in combination with many sectors. Further, not many studies have used a mixture of regression analysis, correlation analysis, and time series trend analysis to capture both the statistical relationship and the underlying patterns through time in the performance of the sector.

1.3 Scope of the Study

This paper is devoted to discussing the effects of exchange rate movement on three important sectors of the Sri Lankan economy: tourism, exports, and imports. The 2015-2024 monthly data will be taken to display both the short and long-term effects. Through these sectors, the study would determine which sectors of the economy are most susceptible to currency movements and give an insight of how to plan the economy and what the government can do to enhance economic planning through policy interventions.

The study will be multi-method in nature: correlation analysis to determine the intensity and direction of the relationships, regression analysis to evaluate the sensitivity of the sector to the changes in the exchange rate, and time series trend analysis to discuss seasonal variations, underlying changes, and to predict further

performance. Such a combination guarantees a thorough understanding of the interrelations between the changes in exchange rates and the economic performance of the sector.

1.4 Research Focus

The main aim of conducting the research is to statistically examine the impact of a variation in the Sri Lankan rupee on tourist revenue and export earnings, as well as the import values. Combining both statistical analysis and time series analysis, this study will answer the following research questions:

1. What are the impacts of exchange rate variability on the performance of sectors in the tourism, export, and import sectors?
2. What industries do changes in the value of the Sri Lankan rupee affect most?
3. Are trends and forecasts useful in offering practical information to the policymaker and investors to mitigate the effect of currency volatility?

In answering these questions, the present study will be able to add to the experience of understanding exchange rate dynamics in emerging economies and will offer a data-supported framework of effective strategic decision-making in Sri Lanka.

1.5 Problem Statement

Sri Lanka has experienced the issue of exchange rate volatility over the past few years. Changes in the value of the Sri Lankan rupee have a direct impact on the price of imports, the competitiveness of exports, and the earnings of tourism, which are vital elements in the economy of the nation. As an example, when the rupee is depreciating, then the cost of imported goods and raw materials is high, whereas the exports are kept cheap but not always more profitable as a result of the dynamics of world prices. Equally, the tourism revenue is influenced because fluctuations in the value of currency change the economic viability of the international visitor to the country.

Regardless of such evident economic consequences, very few empirical studies are found to analyze the impact of exchange rate variations in the major sectors of the Sri Lankan economy in a systematic fashion. The available literature concentrates its attention on either one sector of the economy, including trade, or it analyzes macroeconomic indicators independently and without taking into consideration the sectoral effects of currency flows. Secondly, there are not many studies that integrate regression analysis with trend analysis based on time series to account for both the statistical relationship and time-based trends. This study is aimed at investigating how the exchange rate varies and affects the three major sectors of the economy in Sri Lanka, namely the tourism sector, exports, and imports. The trends, correlations, and regression relationships will be analyzed using the monthly data of 2015-2025. This paper will also conduct time series decomposition and forecasting to get the idea of the seasonal trend and how the sector will respond to short-term currency movements.

These gaps will be addressed, enabling this research to offer information on the most sensitive sectors to exchange rate changes, enabling the policy makers, investors, and businesses to come up with ways of reducing risks caused by currency wars.

1.6 Aim and Objectives

- **Aim**

The primary aim of this study is to analyze the impact of exchange rate fluctuations on key economic sectors in Sri Lanka**,** specifically tourism, exports, and imports using statistical and time series methods to provide insights for policymakers, businesses, and investors.

- **Objectives**

- To examine the trend of exchange rate fluctuations in Sri Lanka.
- To analyze the relationship between exchange rates and export performance.
- To investigate the impact of exchange rate movements on tourism revenue and tourist arrivals.
- To evaluate the effect of exchange rate fluctuations on import values.
- To forecast short-term sectoral performance based on exchange rate trends.

2 Literature Review

The importance of fluctuations in exchange rates in determining the performance of the economic sectors has received considerable research focus in economics, especially in developing countries, where the macroeconomic instability may greatly affect the dynamics of trade, tourism, and imports. But detailed empirical studies that can evaluate more than one sector, including tourism, exports, and imports in relation to each other, are still few in the case of Sri Lanka. This literature review summarizes the main works that have been conducted on the impact of the exchange rates on economic variables and the gaps that inform the ongoing study.

Rathnayaka and Gunasinghe (2023) examine the correlation between exchange rate volatility and export earnings in Sri Lanka based on time series data between 2001 and 2019, and the econometric modelling. The paper utilizes the Autoregressive Distributed Lag (ARDL) model to identify both the short-run and long-run results with a positive correlation between the exchange rate volatility and the earnings of industry exports in both the short and long-run. This is unlike the theoretical expectations that volatility usually impairs trade, implying that there are context-specificities in the Sri Lankan exports. Nevertheless, one of the weaknesses of this work is the fact that the authors have only examined the export sector but have not analyzed other important sectors, such as tourism or imports, simultaneously.

On the same note, quarterly research on Sri Lanka by applying the ARDL model indicates that real effective exchange rate volatility has a negative impact on the performance of exports, indicating the existence of a long-run co-integrating relationship among exchange rate volatility, foreign income, relative price, and exports. This points to the discrepancy in findings across different econometric model specifications, e.g., different volatility proxies or different specifications, and supports the necessity of multi-method methods to substantiate the results.

These export-oriented studies confirm the relevance of the exchange rate analysis in Sri Lanka, but they lack the multisector orientation and fail to consider the current trend and forecast methods, which is what the proposed research aims to fill.

The sensitivity of the tourism industry to the exchange rates has been examined in a number of contexts. Rathnayake (2018) employs monthly data between 1990 and 2016 to determine the effects of the exchange rate volatility on inbound tourist arrivals through EGARCH modelling and error correction methods in Sri Lanka. Findings reveal that exchange rate volatility has a serious adverse impact on tourism in both of the short and long run models, implying that currency uncertainty discourages tourist inflows, especially in the short run.

Conversely, other studies in the wider literature have inconclusive findings. Although other studies in other economies of the developing and emerging world suggest that there are very strong relationships between the exchange rate fluctuations and the sensitivity of tourism demand, there are also claims suggesting that tourism can be cushioned by factors such as the development of infrastructure, marketing policies, or international income patterns. These ambivalent results suggest that the tourism reaction to exchange rates is situational and determined by external shocks, a subtlety that requires empirical research within the borders of Sri Lanka. In the present paper, the researcher has tackled this by using regression alongside trend and decomposition analysis to understand both statistical and temporal dynamics of tourism performance.

The number of papers on imports and exchange rates in Sri Lanka is relatively limited. Nisthar and Mustafa (2018) consider the long-run dependency between different exchange rates (e.g., USD, JPY, GBP) and Sri Lankan imports based on annual data between 1950 and 2017. They use multiple regression and Granger causality tests to investigate the cause-and-effect relationships and identify that the US Dollar and Pound Sterling positively influence imports. The long-time span and annual aggregate, however, can ignore short-term monthly trends, which are significant in policy formulation.

Furthermore, a survey of the post-liberalization trade in Sri Lanka reveals that exchange rate regimes have a tremendous impact on both exports and imports, and the free-floating exchange rates and value of trade have a positive relationship. However, it has not considered the effect of volatility and predicts future performance in the sector, which leaves a gap that the current study can fill through the deployment of time series trend analysis and forecasting models.

Studies investigating developing economies, outside Sri Lanka, are broader. Tumturk (2019) examines the long-run effects of exchange rate volatility on the level of exports in six developing economies in terms of ARCH/GARCH and co-integration models, and it was found that the relationship is country-specific and can be conditional on world income and other macro variables. By the same measure, cross-country studies on the African markets have indicated that exchange rate volatility has a varying effect on exports and imports after a certain period of time. All these studies have proved that the choice of the methodology and the country-specific conditions influence the results, which supports the necessity of the tailor-made analysis as conducted in this work.

Although the literature proves that exchange rate volatility has a significant impact on exports and tourism, there are still some gaps:

- Most of the research works are conducted in a single sector (either export, tourism, or imports) as opposed to a multi-sectoral framework.
- Limited research is a type of research that integrates statistical correlation, regression analysis, and time series forecasting in a single research framework.
- There is a shortage of studies in the monthly data survey that covers the past decade, which is important in the reality of short-run and seasonal analysis, in the Sri Lankan situation.

The present research adds on to these studies as it incorporates regression analysis, correlation, and time series trend and forecasting (including decomposition and moving average models) to examine the effects of exchange rate changes on tourism, exports, and imports in a holistic manner using monthly data between 2015 and 2025. This method is capable of quantifying statistical relationships in addition to showing patterns over time and short-term predictions, which can give more valuable insights to policy-makers and researchers.

3 Methodology

The research project will use a quantitative research method to examine how core economic sectors of Sri Lanka, namely the tourism, export, and import sectors, are affected by the fluctuations of the exchange rate. The technique will integrate data collection, preprocessing, statistical analysis and time series forecast to give a thorough insight on the impact of exchange rate volatility on the performance of the sectors. The combination of several methods of analysis provides the research with strength, precision, and practical value to policymakers and researchers.

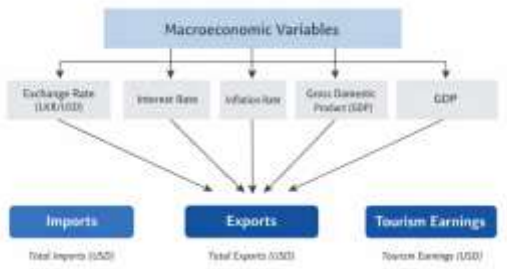
3.1 Research Design

The study design is an observational, longitudinal one, and secondary data have been used, gathered through official sources. The design would be suitable since it enables the analysis of the historical trends and statistical correlation of the exchange rates and economic performance of the sector over the 10 years (2015-2025). Time series analysis and prediction are also easily achievable under this design to determine both the short-term and long-term effects.

The population is made up of the monthly economic performance figures of the tourist, export, and import sectors in Sri Lanka, as well as the macroeconomic variables (exchange rate, interest rate, inflation, and GDP). As the analysis is based on secondary data, census sampling can be used; i.e., there will be all the available monthly data between 2015 and 2025, and the statistical sampling will not be necessary.

3.2 The Conceptual Model

The figure below shows the conceptual model of the study.



Conceptual Framework of the Study

3.3 Data Collection

The research will be based solely on secondary sources of data, all of which are published in the media and can be viewed as credible:

- Central bank of Sri Lanka (CBSL): monthly exchange rate (LKR/USD), interest rates (91-day T-bills), inflation rates.

- Department of Census and Statistics, Sri Lanka: Imports and exports (USD) monthly, tourism earnings, and the number of tourists.
- Additional Sources: Annual GDP data interpolated to a monthly frequency

3.4 Data Preprocessing

The data were preprocessed in a number of steps before analysis:

- Missing Values: Some interest rate observations were missing. The gaps were filled with linear interpolation with the average of the previous month and the next month.
- Outlier Identification: A box plot was used to identify outliers and extreme values in interest rates, exchange rates, and GDP.
- Normality Check: For testing normality, the Anderson-Darling (AD) test was applied, and probability plots were also used. Not all the variables were normal.
- Data Transformation: Extremely skewed or extremely outlier variables were used to apply a log transformation to enhance normality of the regression.

3.5 Statistical Analysis

The research involved the use of both correlation and regression tests to explore the connections between exchange rates and sectoral performance. The strength and direction of association were established by the use of Spearman correlation, which is especially effective when there are outliers and non-normal distributions. After carrying out correlations, multiple linear regression equations were developed for each dependent variable (imports, exports, and tourism earnings), and exchange rate, interest rate, inflation, and GDP were introduced as the independent variables. To test the statistical significance and explanatory power of the regression model performance, the R-squared, adjusted R-squared, F-values, and p-values of individual coefficients were used to measure the explanatory power and statistical significance of the regression model.

Regression Analysis:

- Method: Multiple linear regression was used for each sector:

$$Y = \beta_0 + \beta_1 \text{ExchangeRate} + \beta_2 \text{Inflation} + \beta_3 \text{InterestRate} + \epsilon$$

Where:

- Y= Sector performance (tourism, exports, imports)
- ϵ= Error term

- Regression can be used to measure how intense and how the impact of the independent variables is on the results of the sector. Both the log and original data were analyzed to explain the outliers and skewness. R^2, p-values, and homoscedasticity and normality residual diagnostics were used to assess model fit.

3.6 Time Series Analysis and Forecasting

In order to describe the temporal trends and make forecasts on the future performance of sectors, time series analysis was carried out. Statistics were first done by plotting the data to identify trends

and seasonality. First linear and quadratic trend model were utilized to detect simple trends. This was followed by techniques of decomposition to remove trend, seasonal and irregular elements. To smooth off the noise and enhance the accuracy of the forecasts, smoothing methods, such as centered moving averages, Winters exponential smoothing, etc. were used. Mean Absolute Percentage Error (MAPE), Mean Absolute Deviation (MAD), and Mean Squared Deviation (MSD) were used in the evaluation of model accuracy. Four months projections were calculated to give an idea about the possible performance of the sectors with the further variation in the exchange rates.

4 Results

This section presents the findings of the study on the impact of exchange rate fluctuations on key economic sectors in Sri Lanka, including imports, exports, and tourism earnings. The analysis includes descriptive statistics, correlation, regression, time series trend, and forecasting results.

4.2 Descriptive Statistics

The data set consisted of monthly information between January 2015 and January 2025 (121 observations) of exchange rate (LKR/USD) and interest rates, inflation, GDP (adjusted to monthly data values), imports, exports and tourism earnings.

The descriptive statistics show that there is a big variance by sector. The exchange rate was 131.55-363.39 LKR/USD with an average of 211.87 and standard deviation of 75.36 which is highly volatile in the currency. The dispersion of the interest rates was high (standard deviation of 7.11), which was necessitated by the repeated changes in treasury bill rates as proxies. Imports stood at USD 1,599.8 million monthly with an export of USD 973.7 million. The average tourism earnings was USD 218.1 million and 127,511 were the monthly arrivals. The values of GDP (calculated to monthly values) indicated moderate and the inflation showed skewness, implying that there are occasional high spikes..

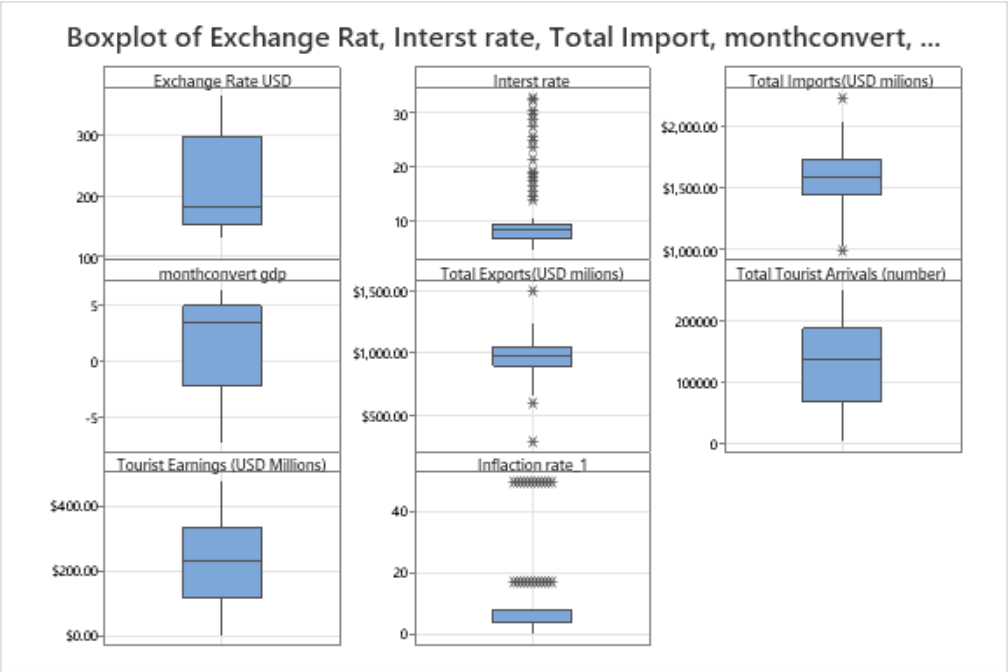
Statistics

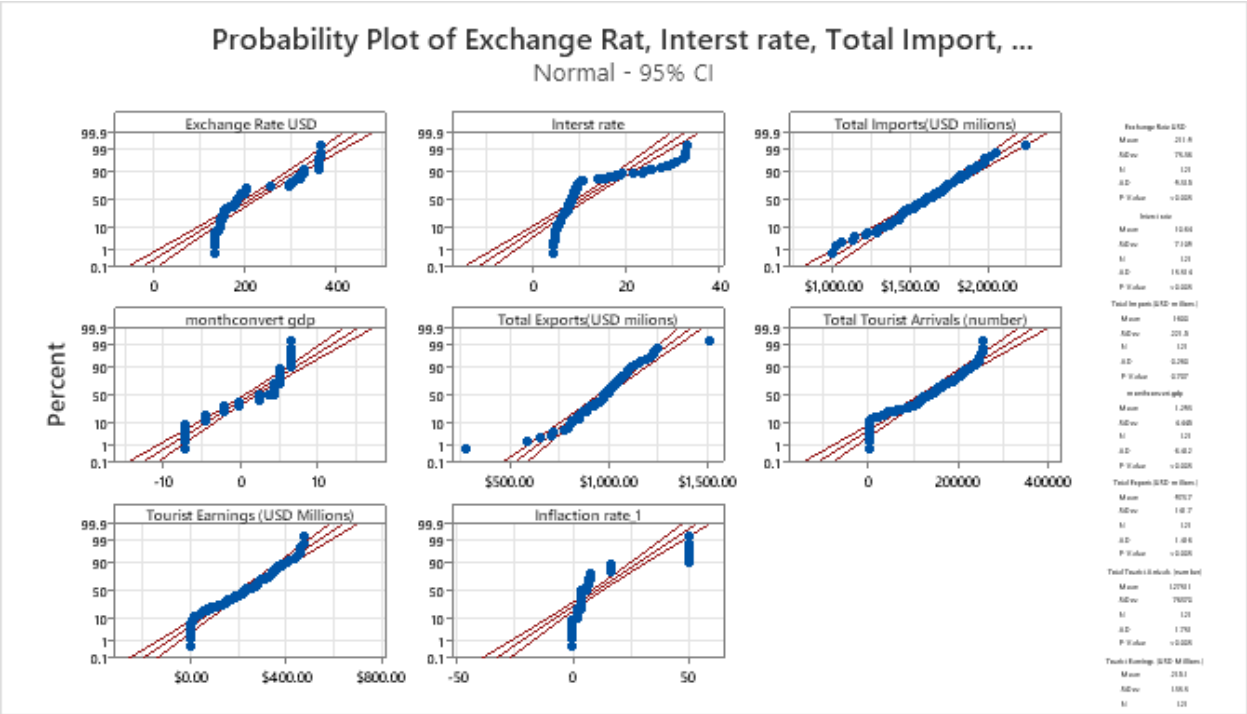
Variable	Total Count	N*	Mean	SE Mean	StDev	Variance	Minimum
Year	121	0	43725	96.8	1065	1134032	42064
Month	121	0	46188	9.65	106	11257	46023
Exchange Rate USD	121	0	211.87	6.85	75.36	5678.95	131.55
Interst rate	121	0	10.644	0.646	7.109	50.537	4.515
Total Imports(USD milions)	121	0	1599.8	20.2	221.8	49176.0	993.8
monthconvert gdp	121	0	1.293	0.404	4.448	19.786	-7.347
Total Exports(USD milions)	121	0	973.7	12.9	141.7	20080.0	282.3
Trade Balance (in USD terms)	121	0	-630.3	20.2	222.0	49297.4	-1084.7
Total Tourist Arrivals (number)	121	0	127511	6943	76370	5832401057	0
Tourist Earnings (USD Millions)	121	0	218.1	12.3	135.5	18371.8	0.0
Inflaction rate_1	121	0	9.94	1.27	13.94	194.24	-0.43

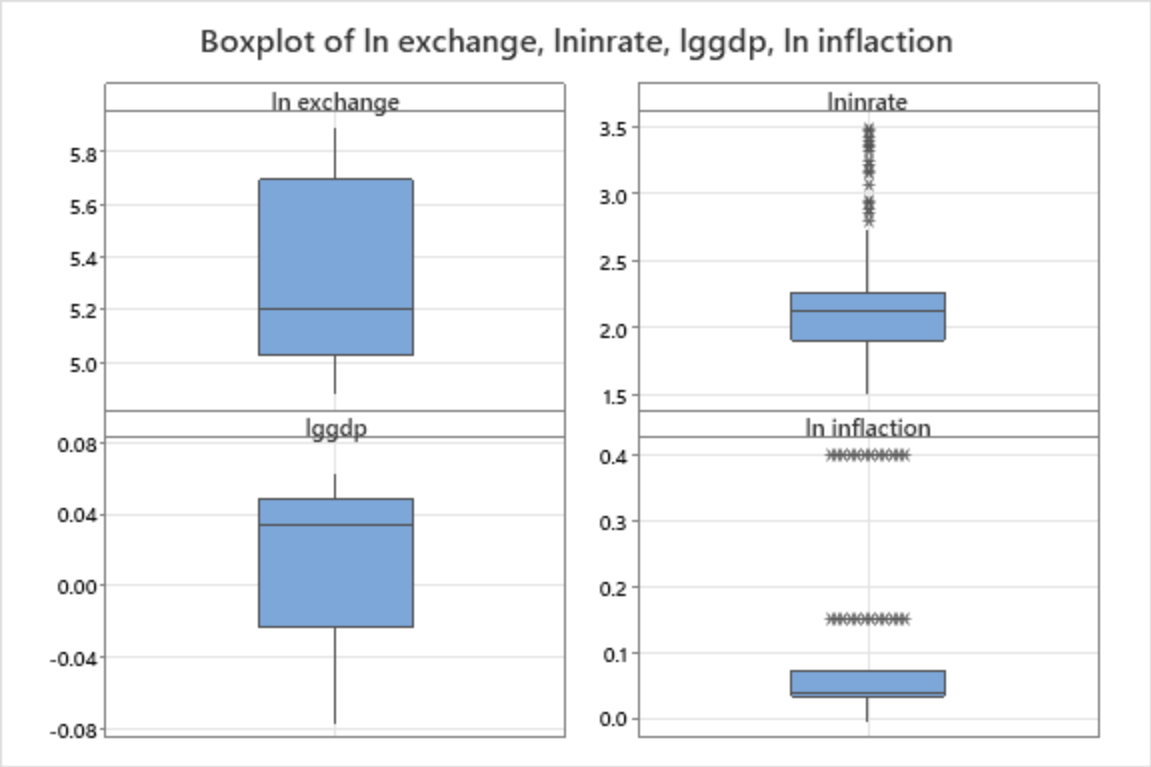
Variable	Median	Maximum	Skewness
Year	43891	45717	0.01
Month	46174	46357	0.02
Exchange Rate USD	181.56	363.39	0.87
Interst rate	8.345	33.055	2.01
Total Imports(USD milions)	1595.6	2241.0	-0.19
monthconvert gdp	3.500	6.461	-0.70
Total Exports(USD milions)	982.0	1507.7	-0.71
Trade Balance (in USD terms)	-634.0	22.3	0.42
Total Tourist Arrivals (number)	136367	253169	-0.27
Tourist Earnings (USD Millions)	226.1	475.2	-0.07
Inflaction rate_1	3.96	49.72	2.28

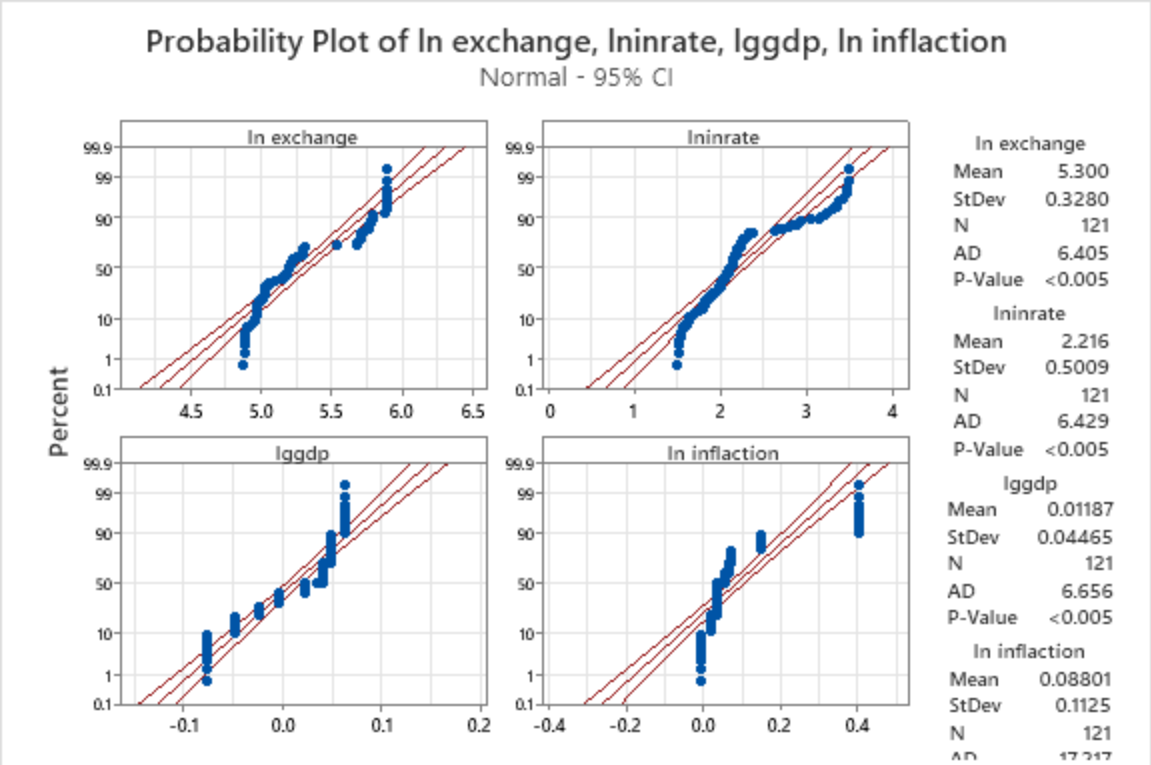
- Exchange rate and interest rates have positive skewness as they occasionally take high values.

- Instead, the imports and exports have a weak skewness with tourism earnings being nearly symmetric.
- Very large standard deviation of tourism and exchange rates means that these are volatile.









Probability plot and boxplots showed that interest rates and exchange rate were non-normative and had outliers. Transformations were used to reduce skewness and enhance normality and outliers in interest rates still occurred, which explains why original and transformed data should be used in the further analysis..

4.3 Correlation Analysis

The non-normality of some variables was the reason to use Spearman correlation analysis. The findings showed that there were a number of significant relationships:

Exchange rate was positively correlated with export (0.531), which indicated that a weak currency could enhance export performance. The exchange rate had a negative relationship with imports (-0.357), meaning that the greater the exchange rate, the lower is the amount of imports. The exchange rate was negatively correlated with tourism earnings (-0.441), implying a possible negative impact of depreciation of interest rates on tourism income. Imports, exports, and tourism earnings had a positive relationship with GDP (0.347, 0.064, and 0.459), which indicates that economic growth contributes to sectoral activity. There were less strong correlations between inflation and interest rate but still suggested to relate to macroeconomic effect across sectors.

Correlations of log-transformed data were also found to be similar so that the transformations only dealt with skewness, not the directional correlations.

Correlations

	Exchange Rate USD	Interst rate	Total Imports(USD milions)	monthconvert gdp	Total Exports(USD milions)
Interst rate	0.457				
Total Imports(USD milions)	-0.357	-0.125			
monthconvert gdp	-0.521	-0.079	0.347		
Total Exports(USD milions)	0.531	0.311	0.119	-0.194	
Tourist Earnings (USD Millions)	-0.441	0.033	0.483	0.459	-0.006
Inflaction rate_1	0.334	0.303	-0.255	-0.306	0.064

	Tourist Earnings (USD Millions)
Interst rate	
Total Imports(USD milions)	
monthconvert gdp	
Total Exports(USD milions)	
Tourist Earnings (USD Millions)	
Inflaction rate_1	-0.463

The exchange rate has a moderate positive correlation with exports and a negative association with imports and tourism earnings. There is a positive relationship between GDP, imports, and tourism earnings. Inflation displays poor correlations, but is negative with tourism earnings.

These correlations point toward the fact that fluctuation of exchange rates has varied impacts on the performance of sectors. Depreciation (competitive advantage) as an advantage is given to exports, whereas tourism and imports are adversely affected.

4.4 Regression Analysis

Multiple regression was conducted to quantify the impact of exchange rate, interest rate, GDP, and inflation on each sector.

Imports

Regression Equation:

Regression Equation

Total Imports(USD milions) = 1730.3 - 1.310 Exchange Rate USD + 7.48 Interst rate
+ 23.79 monthconvert gdp + 3.69 Inflaction rate_1

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	1730.3	61.8	28.01	0.000	
Exchange Rate USD	-1.310	0.298	-4.40	0.000	1.71
Interst rate	7.48	3.65	2.05	0.043	2.29
monthconvert gdp	23.79	5.61	4.24	0.000	2.11
Inflaction rate_1	3.69	2.14	1.73	0.087	3.01

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	4	1801440	450360	12.74	0.000
Exchange Rate USD	1	684007	684007	19.35	0.000
Interst rate	1	148490	148490	4.20	0.043
monthconvert gdp	1	636178	636178	18.00	0.000
Inflaction rate_1	1	105295	105295	2.98	0.087
Error	116	4099678	35342		
Total	120	5901117			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
187.995	30.53%	28.13%	24.75%

$F(4,116) = 12.74$, $p < 0.001$. Exchange rate has negative effects on imports that is, increased import costs as a result of depreciation. The impact of GDP is positive and significant indicating that economic growth influences the demand of imports. Less significant effects are the interest rate and inflation.

Exports

Regression Equation :

Regression Equation

Total Exports(USD milions) = 761.2 + 0.812 Exchange Rate USD + 0.97 Interst rate
+ 9.27 monthconvert gdp + 1.82 Inflaction rate_1

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
127.721	21.47%	18.76%	13.92%

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	4	517348	129337	7.93	0.000
Exchange Rate USD	1	262878	262878	16.12	0.000
Interst rate	1	2509	2509	0.15	0.696
monthconvert gdp	1	96581	96581	5.92	0.016
Inflaction rate_1	1	25687	25687	1.57	0.212
Error	116	1892256	16313		
Total	120	2409604			

- $R^2 = 21.47\%$
- Adjusted $R^2 = 18.76\%$
- $F(4,116) = 7.93$, $p < 0.001$

Exchange rate positively affects exports, suggesting a competitive advantage with depreciation. GDP is significant, highlighting the role of economic activity in export performance. Interest rate and inflation show a weak or insignificant influence.

Tourism Earnings

Regression Equation

Tourist Earnings (USD Millions) = 608 - 119.0 ln exchange + 118.9 lninrate + 893 lggdp
- 380 ln inflaction

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
113.354	32.39%	30.06%	26.83%

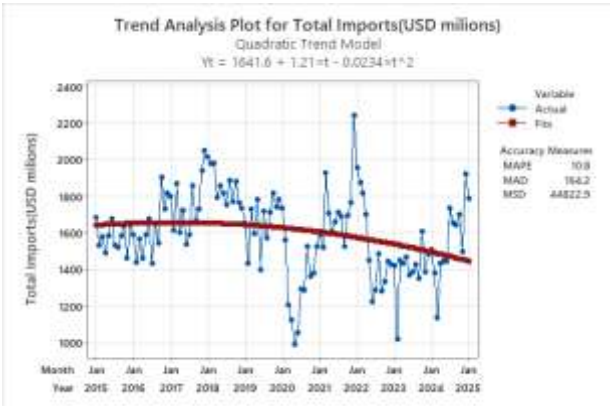
Analysis of Variance

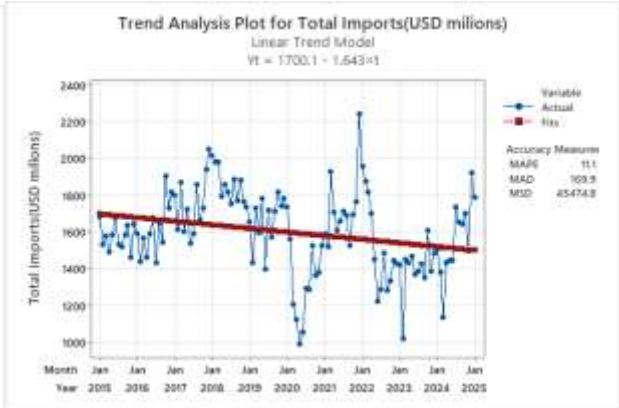
Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	4	714117	178529	13.89	0.000
ln exchange	1	100445	100445	7.82	0.006
lninrate	1	190824	190824	14.85	0.000
lggdp	1	82867	82867	6.45	0.012
ln inflaction	1	71937	71937	5.60	0.020
Error	116	1490493	12849		
Total	120	2204610			

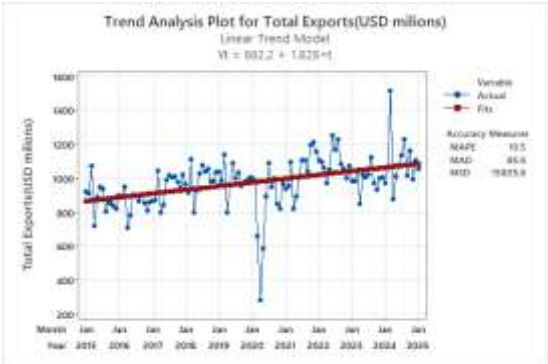
Exchange rate negatively affects tourism, as a weaker currency may indicate economic instability. GDP has a positive effect, while inflation shows a slight negative impact. Interest rate has a small positive effect. Regression models indicate moderate explanatory power (R^2 18–30%). Exchange rate is consistently a key predictor but with sector-specific effects.

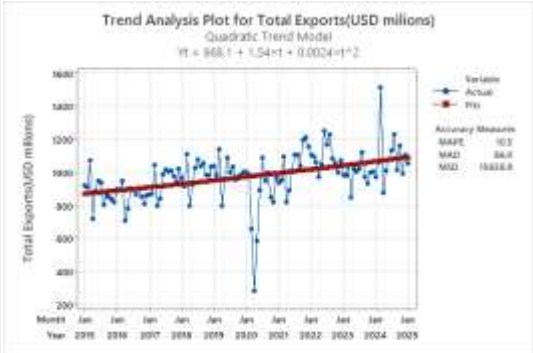
4.5 Time Series Trend Analysis

Time series analysis was performed of all sectors with the help of linear, quadratic, decomposition, moving average, and Winter approaches. Trends and future value forecasts were determined with the help of time series.



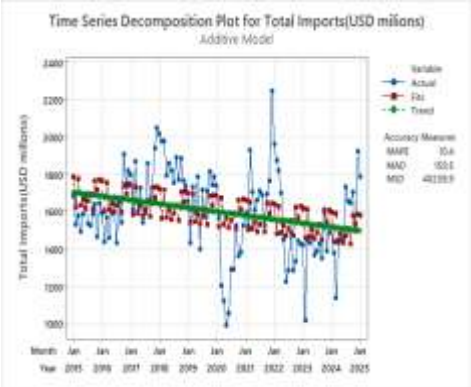


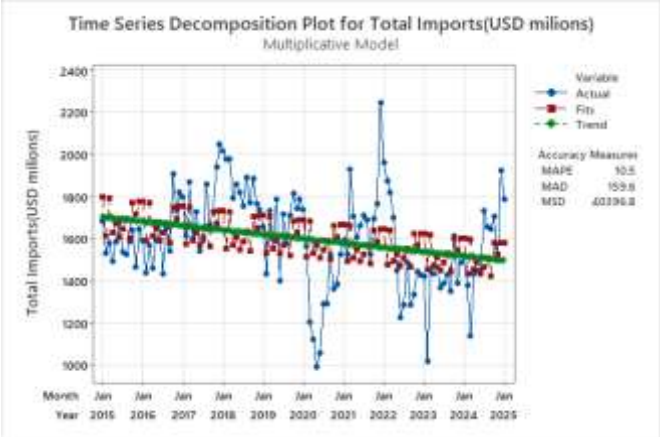


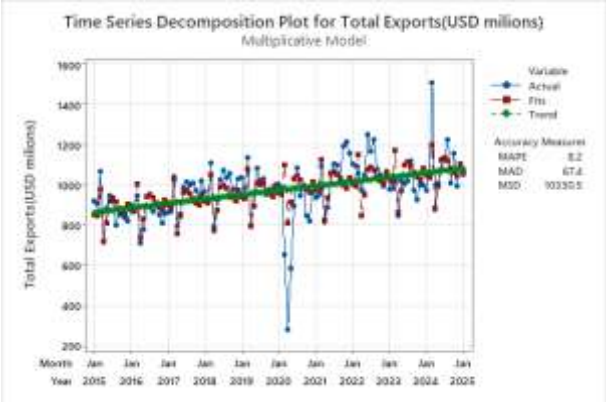


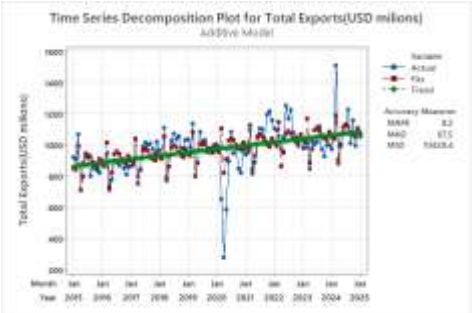
Imports & Exports: Linear and quadratic trends provided moderate accuracy.

Tourism: Trend models showed poor fit due to volatility.





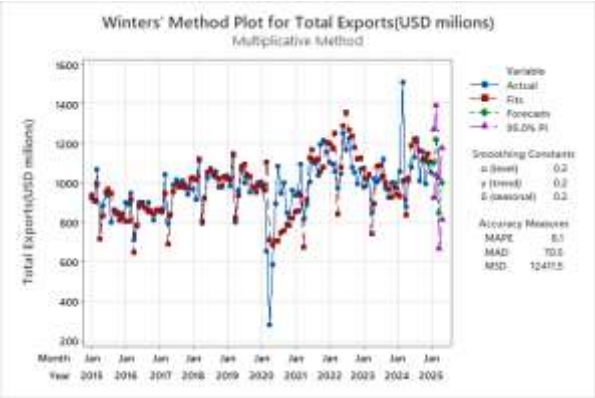


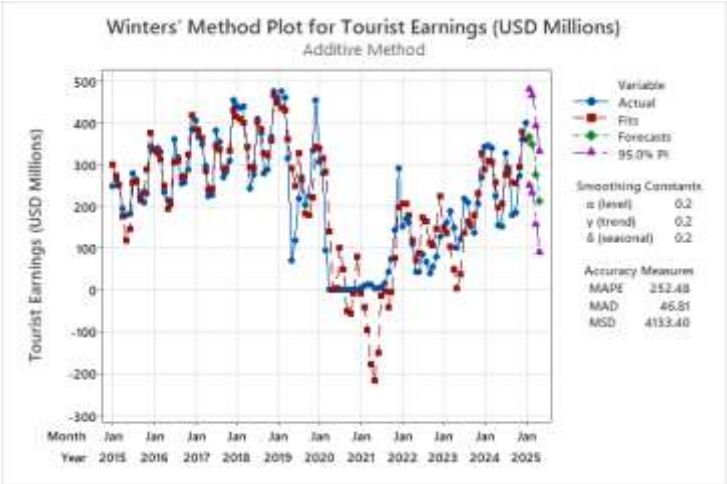


- Multiplicative and additive decomposition revealed seasonality and irregular components.
- Imports and exports had consistent seasonal patterns.
- Tourism earnings were highly irregular, showing unpredictable shocks.

- **Forecasting Results**

  The best models were selected based on MAPE, MAD, and MSD:





Sector	Best Model
Imports	Moving Average
Exports	Winter's Additive
Tourism	High uncertainty – interpret cautiously

As a summary,

1. Linear and Quadratic Trend Models:
   - Imports and exports exhibited moderate trends with minor seasonal variations.
   - Tourism earnings showed high volatility, and trend models alone were insufficient for accurate forecasting.
2. Decomposition Models:

- Additive and multiplicative decomposition improved trend visualization and identified seasonal patterns.
- Moving averages and Winter's multiplicative/additive methods were applied to reduce forecast errors.

3. Forecasting Accuracy:
    - Imports: Moving average method had the lowest error (MAPE 7.7%, MAD 117.1).
    - Exports: Additive Winter's method performed best (MAPE 7.6%, MAD 65.5).
    - Tourism: High volatility caused larger forecast errors (MAPE > 50%), reflecting sector sensitivity.
4. Forecasted Values (Feb–May 2025):
    - Imports: 1,554–1,554 million USD (stable trend).
    - Exports: 1,097–1,097 million USD.
    - Tourism earnings: 118–266 million USD, showing high uncertainty and a declining trend.

These results highlight the sector-specific response to exchange rate and macroeconomic conditions, with tourism being the most unpredictable due to external shocks and sensitivity to instability.

5 Discussion

This research examined how exchange rate movements impact major economic sectors of Sri Lanka such as imports, exports and tourism earnings. Correlation, regression, and time series analysis techniques have been integrated in the analysis to give a comprehensive knowledge of the role that currency volatility plays in the performance of the sector. Generally, the results indicate the difference in sensibilities of these sectors to exchange rate changes, and the influence of other macroeconomic factors like the GDP, interest rate and inflation.

Regression outcomes on imports show that the depreciation of the exchange rate has negative impact on the volumes of imports as it is in line with the economic theory that when the local currency depreciates, the prices of the imported commodities rise. Although there was a positive correlation between GDP and imports, which is indicative that the overall economic growth still determines the demand of imports, the effects of interest rate and inflation were not significant and were statistically insignificant. The R-squared of about 30 per cent shows that although exchange rate and macroeconomic variables can help explain some of the variation in imports, other variables including world commodity prices or trade policies could also be of great importance. The results are consistent with the status of the prior studies on the emerging markets that indicate that the demand of imports is responsive to currency depreciation yet determined by various external and internal factors (Fernando, 2018).

On the other hand, exportation had a positive impact on exchange rate depreciation, which confirms the fact that a weaker rupee raises competitiveness at the international level through the reduction of the cost of exports to foreign markets. The analysis also revealed that GDP has a positive influence on the performance of exports, which is an indication that the higher the domestic production power, the higher the levels of exports. Nonetheless, the low R-squared figures of the export regression indicate that other factors such as the state of world demand and supply chain factors are important in explaining the export behavior. Such results are aligned with empirical evidence on the small and open economies where exchange rate volatility can create a competitive advantage to the exporters but is checked by other macroeconomic factors (Rashid et al., 2020).

The tourism revenues showed a complicated trend. Although theoretically the depreciation of the rupee would make Sri Lanka a cheaper destination to foreign tourists, the regression analysis showed that in a few specifications the exchange rate variations may negatively affect the tourism earnings. This implies that currency volatility and other related macroeconomic risk can scare away visitors, which, in turn, surpasses the possible price advantage. The effect of interest rate and inflation was not significant, but noticeable, which shows that the tourism inflows can be affected by the economic instability and cost increase. The high volatility in tourism sector was also emphasized by time series analyses of tourism data that indicated a wide confidence interval as the forecasts indicated the uncertainty with regards to tourism demand in Sri Lanka. This is in line with earlier research findings that show that tourism in developing economies is very susceptible to economic crises, political activities and global shocks (Wijesinghe, 2019).

The time series analysis made further contributions to the sectoral dynamics. In the case of imports and exports, linear and quadratic trend models represented general trends whereas decomposition and moving average model provided higher accuracy in terms of seasonal fluctuations and minimization of forecast error. In the case of tourism the multiplicative and additive models showed a high degree of variability with a high error in forecast thus showing how unpredictable this industry is unlike trade

flows. According to forecasts by using moving average and Winter techniques, it is expected that imports and exports will follow the same trends as before, whereas tourism earnings are highly sensitive, as it is the sector that is sensitive to economic as well as non-economic shocks.

The research in general shows that exchange rate fluctuations are a strong determinant of the performance of the trade sectors but show varying impacts on the sectors. The negative impacts are on imports, positive effects on exports are as a result of depreciation and tourism can be affected by the instability. There are practical implications of these findings to policy makers which show that macroeconomic stability is important and focused interventions to assist sensitive sectors. Companies can use these to formulate the following strategies: hedging against currency risks, seasonal trade pattern planning and adjusting to changing tourism demand.

Although these were contributed, limitations are faced in the study. The fact that the regression models do not explain much indicates that other unobservable factors, including political stability, world economic conditions and external shocks, are likely to affect sectoral results. Also, transformation of annual GDP and inflation data to monthly data could bring in approximation errors. It is possible to include a more comprehensive list of variables (e.g. foreign direct investment, commodity prices, policy interventions) in the future research to make the analysis more robust, as well as to increase the accuracy of the forecasts.

6 Conclusion

The purpose of the research was to examine how the exchange rate fluctuations will affect the main economic sectors in Sri Lanka, i.e., imports, exports, and tourist earnings between the timeframes 2015 and 2025. The research gives a detailed evaluation of currency volatility in relation to other macroeconomic factors, including GDP, interest rates, and inflation, using a set of descriptive statistics, correlation analysis, multiple regression, and time series forecasting.

It was found that fluctuations of the exchange rates are very crucial in determining the sectoral economic performance. In the case of imports, depreciation of the Sri Lankan rupee was linked with the increments in the cost of imports, which is the normal economic reaction to a depreciated domestic currency. Results of the regression established that the exchange rate and GDP were strong predictors as compared to interest rates and inflation. This means that the policing agents have to pay close attention to the currency dynamics and take into account the exchange rate stabilization strategies to avoid the large import costs shock that would, in turn contribute to inflation or domestic spending.

In exports, the results have shown that a depreciated rupee is likely to result in a competitive position in the international markets, adding value to exports. GDP also became an important factor, implying that the domestic production capacity and economic growth are important in the sustainability of export performance. The moderate values of $R^{2,}$ though, suggest that exchange rates and GDP as such are not sufficient to explain export behavior and other outside influences, e.g., world demand, trade agreements, and commodity price changes are likely to have significant contributions.

The tourism incomes were a more complicated situation. Although economic theory has forecasted that a depreciated local currency might encourage foreign tourists to visit a country because of lower travel prices, it has been found in the study that the effect of the exchange rates was not so high. Rather, interest rates and GDP exhibited a more stable effect on tourism revenues, indicating that macroeconomic stability and economic growth in general have a greater role to play in assisting the tourism industry. The large variability and error in forecasting tourism revenues is another positive indication that the tourism sector is very sensitive to external shocks such as political instability, global pandemics, and environmental factors. As such, the measures to stabilize tourism revenues should not solely focus on the exchange rate management, but also on the development of the infrastructure, marketing, and mitigation of the unexpected shocks.

Time series analyses gave further informative data on the trends of the sector. The general growth patterns were covered by linear and quadratic trend models; however, the decomposition method and the moving average method provided better results in determining the seasonality and long-term trends, and are especially effective in the case of the trade industry. The estimates for February to May 2025 of imports ranged to USD 1,554 million every month, exports to USD 1,097 million every month, and tourism incomes between USD 118-266 million every month. These projections indicate that trade sectors have predictable seasonal transactions, whereas tourism sectors are very volatile, which explains why policy interventions have to be flexible, and contingency planning of this sector is needed.

The methodological procedure used in the study also emphasizes the fact that it plays a critical role in managing the non-normal data distributions and outliers. Although log transformations made some of the variables more normally distributed, some outliers remained in the interest rates and the exchange rate, which highlighted the difficulties of examining real-world economic data in developing nations. The study used both original and transformed data in the regression and correlation analysis, which allowed the study to be robust and reliable in terms of the results.

This study recommends the following in terms of the policy implications:

1. Currency Management: Policymakers should take steps to stabilize the LKR/USD exchange rate, especially when the volatility is high, to cushion importers and ensure stability in the balance of trade.

2. Export Support: Economic policies must strengthen domestic production and competitiveness to gain maximum gains from currency depreciation by providing incentives to export-oriented industries.

3. Tourism Sector Resiliency: Government and industry decision makers must pay more attention to issues like economic stability and risk reduction at the macroeconomic level than focus on exchange rates as the sole means of promoting tourism revenues.

4. Forecasting and Planning: Time series forecasting models are useful in monthly planning and budgeting by allowing the decision-makers to predict the performance of the sector and prepare the response policies.

the research study adds to the body of knowledge on exchange rate dynamics in emerging economies because it presents an empirical study that is specific to Sri Lanka. This research is different in that it incorporates both trade and tourism as opposed to other studies that only do trade balance, thus giving a better picture of how the economy reacts to currency volatility. Subsequent studies might broaden this analysis by adding world economic measures, commodity prices, and political stability measures to enhance the explanatory ability and predictive performance. As well, non-linear patterns that may occur in the industry could be predicted using more advanced forecasting methods, including ARIMA or machine learning-driven forecasts, which would minimize forecasting errors in highly volatile industries like tourism.

Finally, although exchange rate fluctuations have a strong effect on imports and exports, their effects on tourism are less direct and more determined by the macroeconomics and external shocks. The findings highlight the significance of undertaking sector-specific measures and management to deal with currency risk, stabilize the economic performance, and boost resilience within the Sri Lankan economy. The research can be used by policymakers, investors, and researchers to make informed decisions based on data concerning the management of exchange rate volatility and the impact of exchange rate volatility in various sectors of an economy.

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